

Hello Learners, this guide is created to help you get started with all setups for the coding module. If you are someone who has technical know-how in Python, please feel free to skip the details. For others, please follow the guidelines one by one.

We will do a 3-step setup to get started:

- 1: Download an interface (called as IDE) in which we can write the code (In our case, it is Pycharm)
- 2: Create a virtual environment for Python
- 3: Setup the libraries that will be needed in the code

So, let's get started....

Step 1: Install PyCharm:

In order to write our code and run it, we need an IDE or Integrated Development Environment. IDE is a software application for programmers. You can use it to write, edit, and organize code more easily. Plus there are additional features for debugging, syntax check, library installation etc.

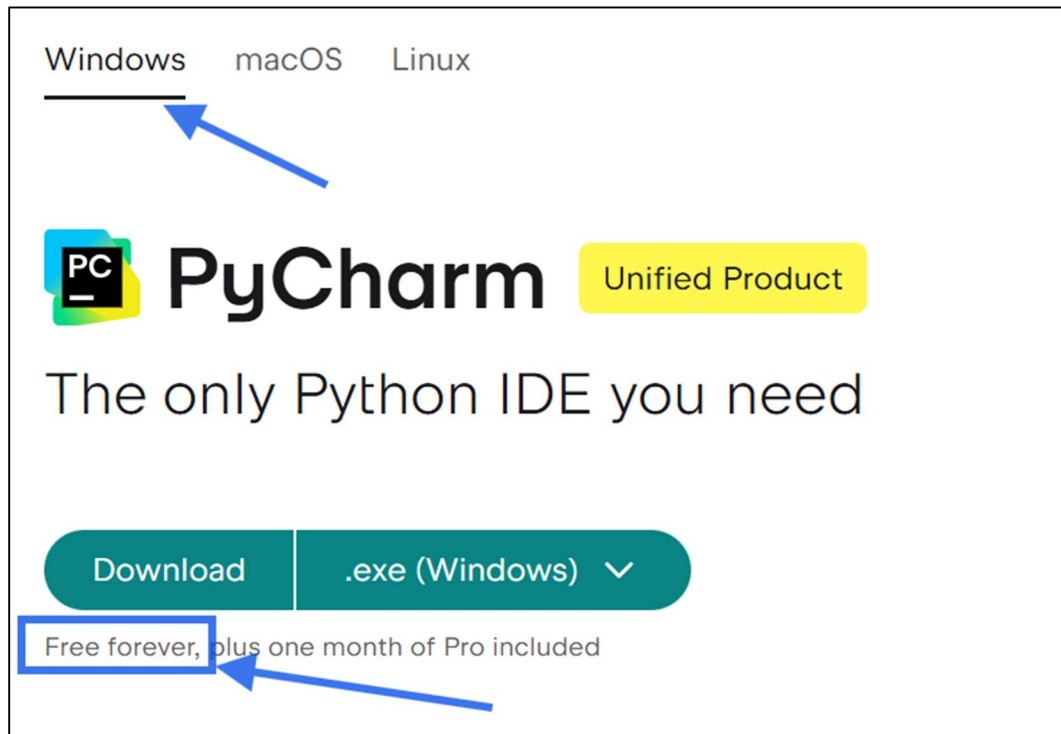
There are many IDEs available in market, each suited to a coding language. For Python, my recommendation is to install PyCharm, which is a nice IDE with host of features. To do so,

a: Google for “download pycharm”

b: Look for the result from “jetbrains” which is the parent company of PyCharm



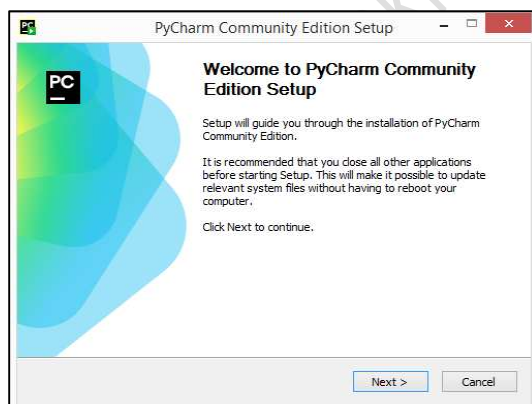
c: Look for the free version



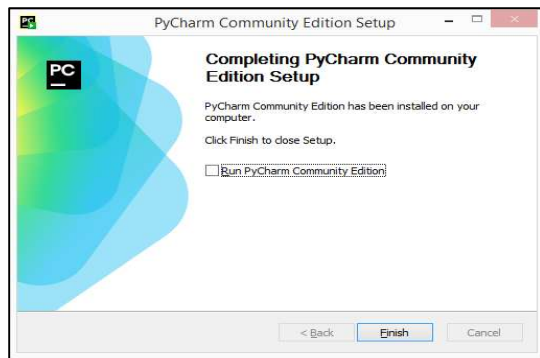
d: Run the downloaded exe file. If you are unsure of your permissions, it's best to run as an administrator

e: Keep clicking "Next". *You do not need to change any option*

It will take couple of minutes for the installation to complete



f: Once the installation is completed, you will get below screen. This means you are all set.



g: Click on Finish.

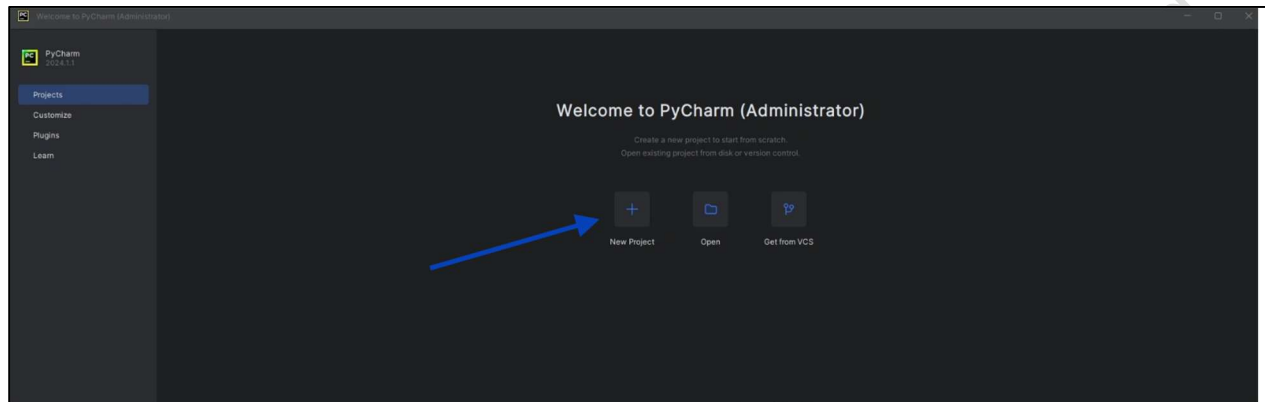
Search for “PyCharm” in windows search to launch it. First time, it may take few minutes to get everything ready.

This concludes our installation of PyCharm

[\[NOTE – We are going to do Steps 2 & 3 in the “Create the chatbot \(Part 1\)” lecture. So, you can skip them for now and come back only if you missed something in the lecture video\]](#)

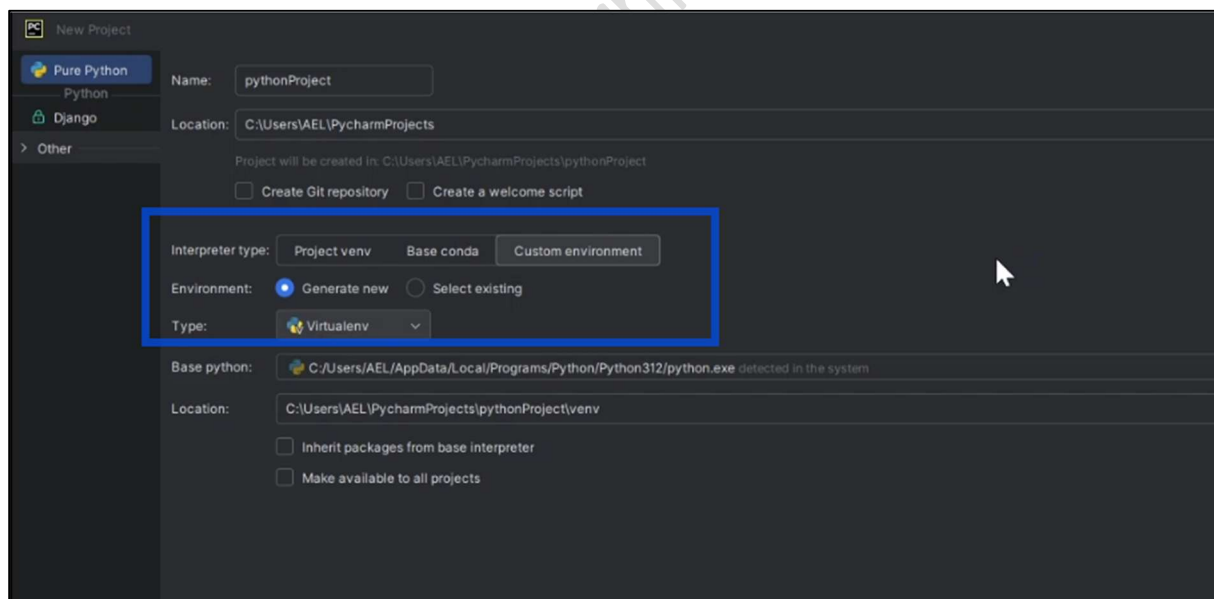
Step 2: Create a virtual environment:

a: If it's the first time you are opening Pycharm, you will see a screen like below. Click on “New Project” (If you already had Pycharm installed, click on File >> New Project)

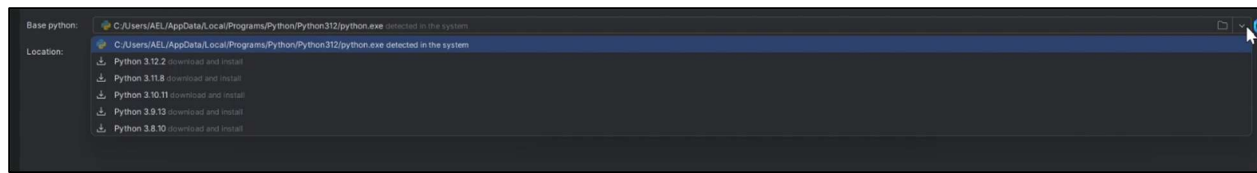


b: You can provide a name and project location of your choice

c: Please ensure you keep the fields in blue box exactly as shown in the screenshot below:

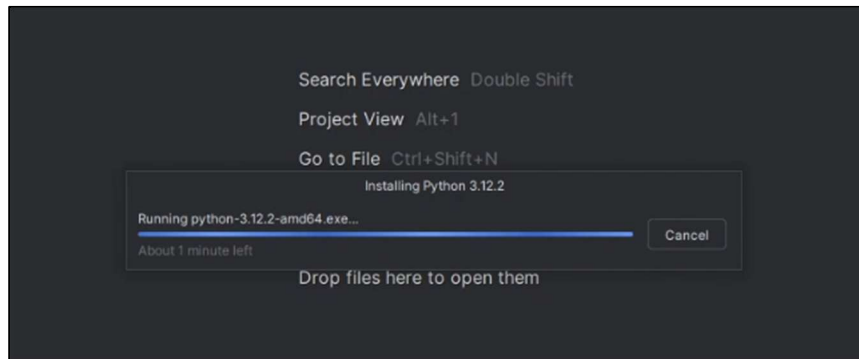


d: Choose a Python version from the dropdown – you can choose the latest version if stable, or one version older to it

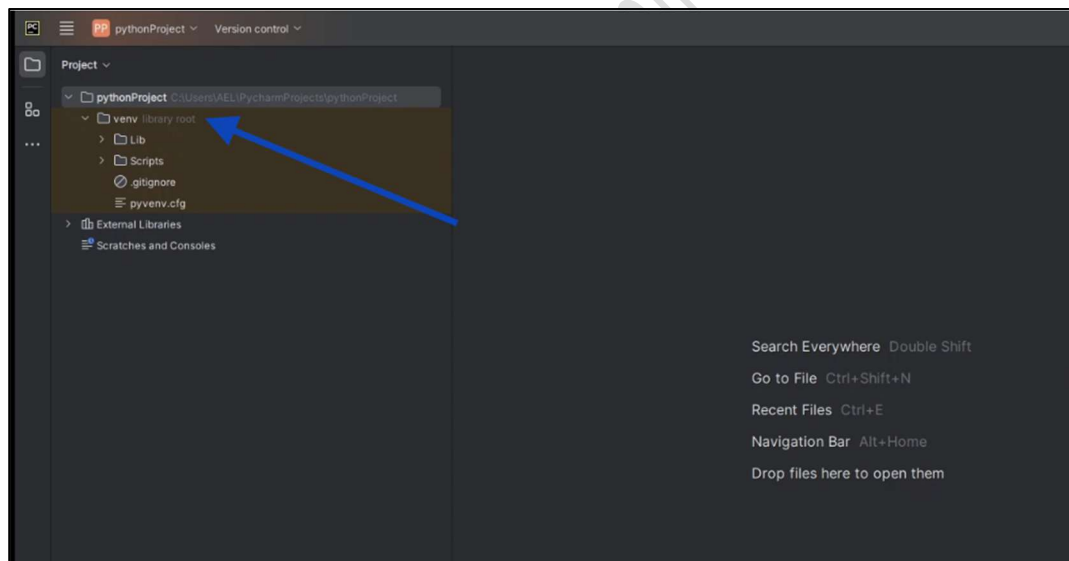


e: Once done, click on the “Create” button

f: This will take a few minutes as you see Python getting installed



g: Once completed, you will land on the explorer screen, where it will show “venv” in the path. This proves your virtual environment setup is completed successfully

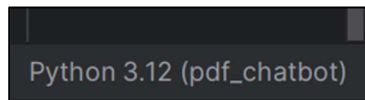


This concludes our setup of virtual environment, including Python installation.

Step 3: Install Libraries (third party packages):

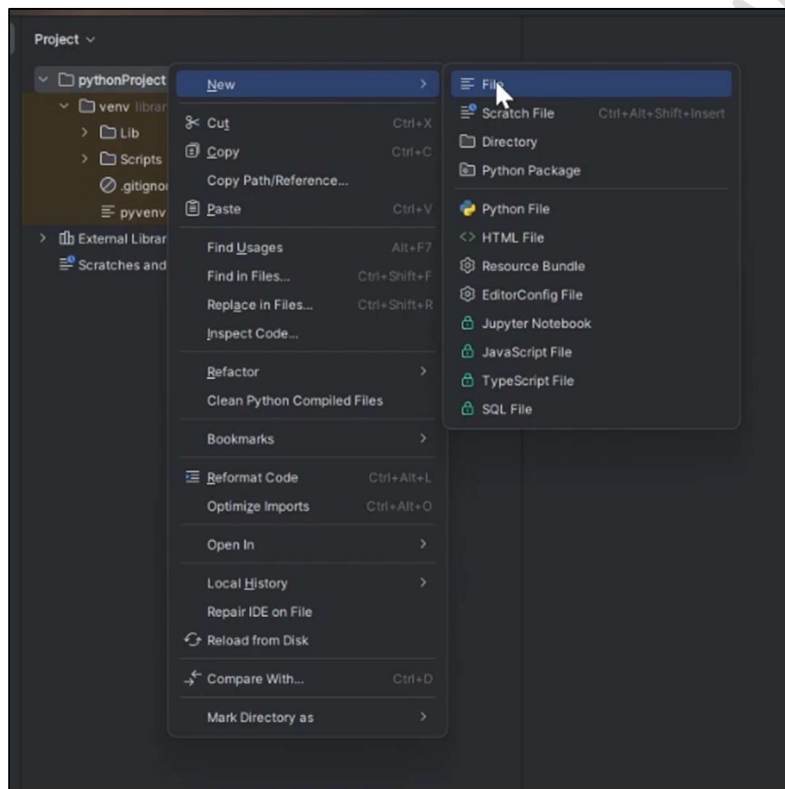
Now that we have installed PyCharm and created a virtual environment, we are all set for the last step, which is installing libraries. Think of libraries as add-ons, which help you do tasks easily. For example, to read the pdf file, we will download a library called pdfplumber which provides several enhanced options to work with pdfs.

Before we install any libraries, it's important to make sure you are working on the right Python version. Look at the bottom right of PyCharm, and make sure it shows the installed Python version.

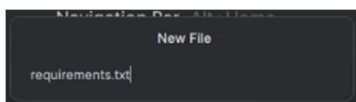


Now we are all set to install the libraries. So let's start:

a: Right click on your project structure, where it says “pythonProject” and choose New >> File



b: Name this file as “requirements.txt”



c: Once the file opens, paste the following library name and versions in the file

streamlit==1.52.2

pdfplumber==0.11.8

langchain==1.2.0

langchain-openai==1.1.6

langchain-community==0.4.1

langchain-core==1.2.5

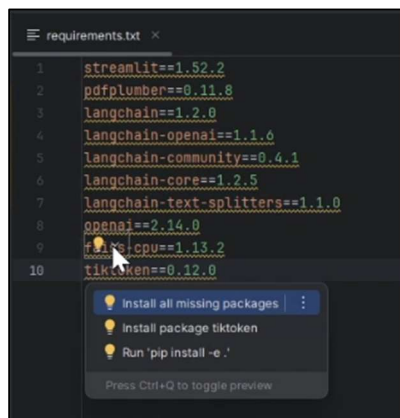
langchain-text-splitters==1.1.0

openai==2.14.0

faiss-cpu==1.13.2

tiktoken==0.12.0

d: You will see a yellow line underneath each library name, because they are not installed. Click on the bulb icon next to any library and click on “Install all missing packages”



e: This is going to take some time (3-5 minutes). Please be patient until all installation is completed

f: Once done you can go to Terminal option, on bottom left, write “pip list” and hit Enter to see the packages that are installed. It should include the above packages, and other dependent packages too.

The screenshot shows the PyCharm IDE interface. The top pane displays the `requirements.txt` file with the following content:

```
1 streamlit==1.52.2
2 pdfplumber==0.11.0
3 langchain==1.2.0
4 langchain-openai==1.1.0
5 langchain-community==0.4.1
6 langchain-core==1.2.5
7 langchain-text-splitters==1.1.0
8 openai==2.14.0
9 faiss-cpu==1.13.2
10 tiktoken==0.12.0
```

The bottom pane shows the terminal output of the `pip list` command. A yellow arrow points to the command prompt, and another yellow arrow points to the `dataclasses-json` package in the list.

```
(venv) PS C:\Users\AEL\PycharmProjects\pythonProject> pip list
Package            Version
-----
aiohappyeyeballs   2.0.1
aiohttp            3.13.3
aiosignal          1.4.0
altair             6.0.0
annotated-types    0.7.0
anyio              4.12.1
attrs              25.4.0
blinker            1.9.0
cachetools         6.2.5
certifi            2026.1.4
cffi               2.0.0
charset-normalizer 3.4.4
click              8.3.1
colorama           0.4.6
cryptography       46.0.3
dataclasses-json   0.6.7
distro             1.9.0
faiss-cpu          1.13.2
frozenlist         1.8.0
gitdb              4.0.12
```

This concludes our installation of third party libraries

Remember if you faced any issue and wanted to uninstall Python or PyCharm you can do so from Control Panel. Make sure to restart your computer post that.

If you face any issue in the above setups, please use the Q&A option to report your issue with the steps, error message and screenshot and we will definitely help you out.

Now we are all set to start coding in the next modules. Good Luck & Keep Learning!!